Group assignment

Business & Marketing Planning

Bilge Berfin Imren
Miray Dizer
Areti Ferderigou
Meili Zhang

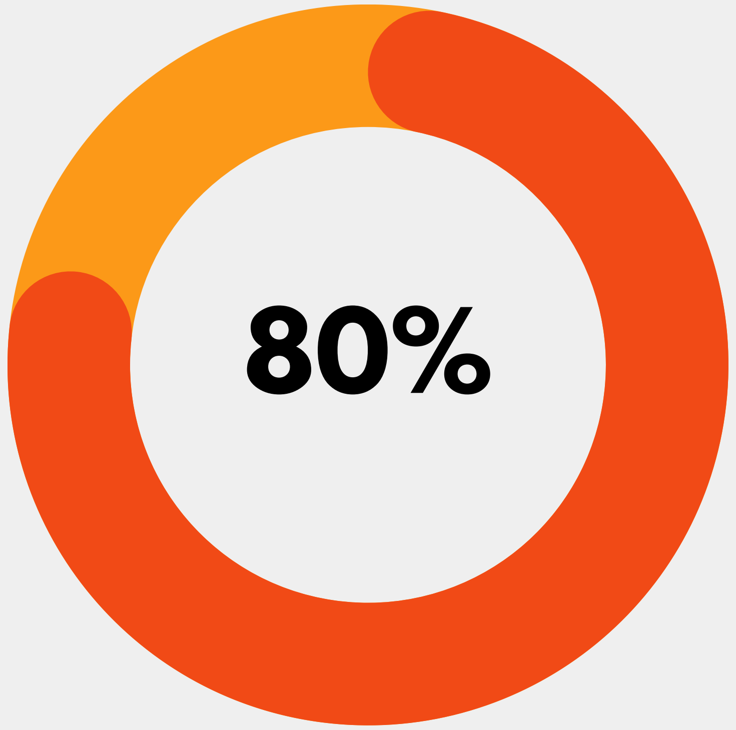
Presentation Agenda

Overview
External analysis
Internal analysis
Weaknesses
Solutions

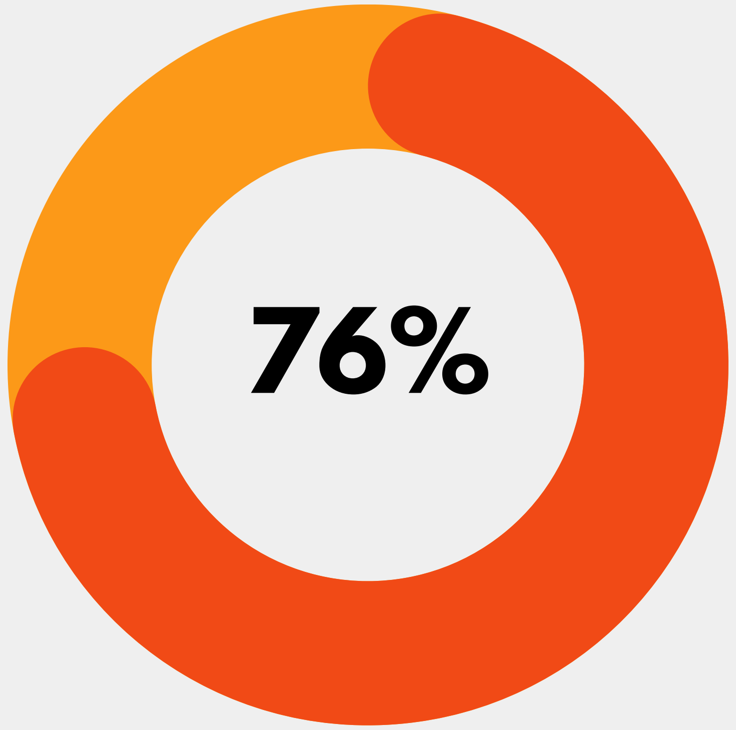
Company Overview

- Sense is utility energy monitoring company selling whole-home monitors and an app.
- It is moving from research grade prototypes toward production grade appliance detection and anomaly-detection features for both B2C and B2B customers.
- Sense's annual revenue to be $44.5M.
- Sense products:
  - Hardware and subscription app features
  - Licenses or partners with utilities and enterprise customers
  - Leverages partnerships to scale distribution and credibility.

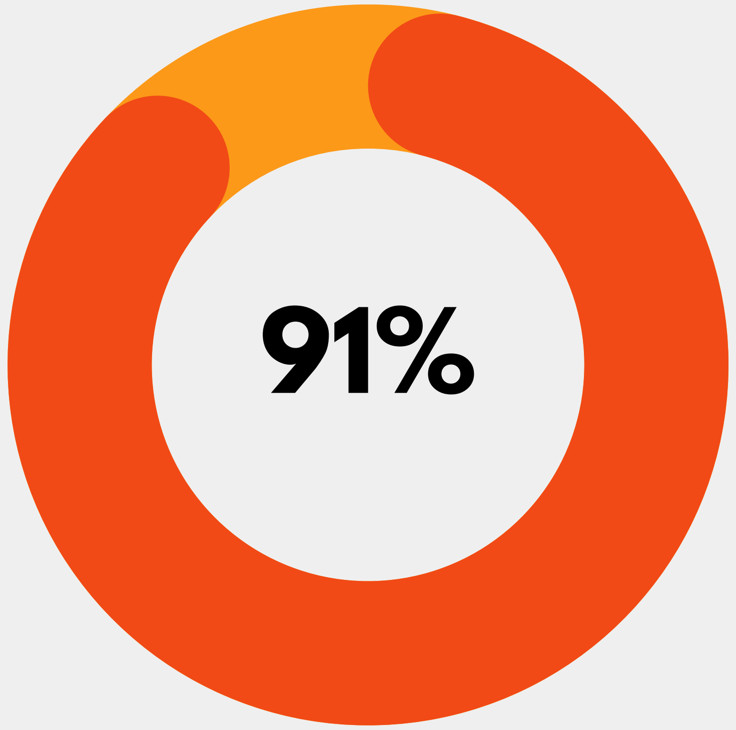
User Statistics of Sense



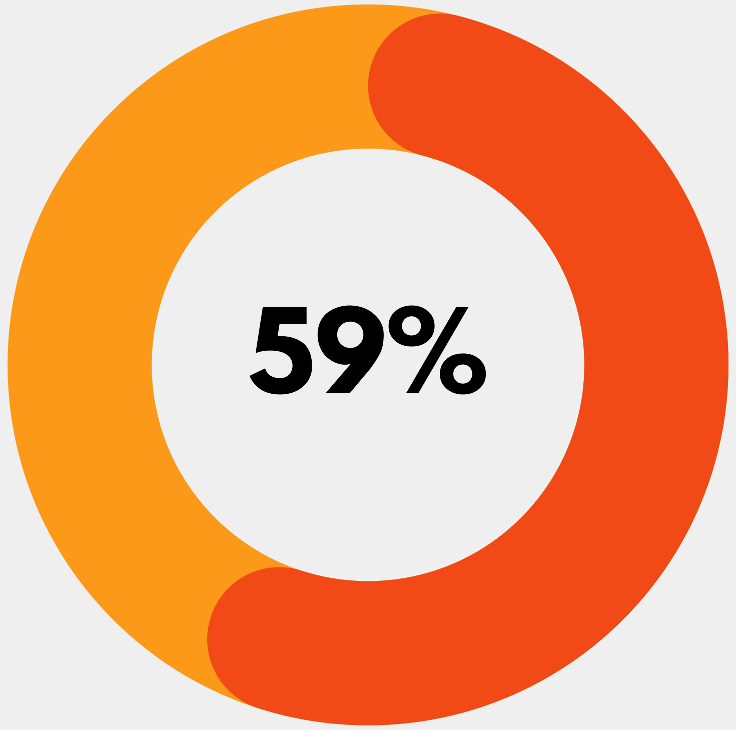
Claimed that they put effort reducing the electricity amount they use



Expressed concerns about cost of energy



Willing to make small changes to help reduce the effects of the global climate change



Made a major investment to improve the energy efficiency house

Market Analysis

PESTLE . 5 FORCES . VALUE CHAIN .
POSITION MAP . VRIO. SWOT. TOWS

PESTLE Analysis

political

- The energy saving and smart grid programs align perfectly with the Sense application.
- Supportive for the growth of Sense

economic

- Investments in green technologies are increasing
- Energy price volatility. Increase of the energy costs push the customer and businesses to track their consumption closely.

social

- The environmental consciousness is increasing, which is in favor of Sense.
- Smart home adoption is related to the easier integration of IoT devices.
- People are more aware and sensitive about our planet and the environment.

technological

- Advances in IoT protocols.
- Adaptation of AI and machine learning

legal

- Data privacy/protection
- IP protection requirements
- Contracts with utilities (B2B)
- Consumer protection laws
- Product safety and certification

environmental

- Climate change awareness in favor of Sense.
- Transition to sustainable energy

FIVE FORCES

competitive rivarly

- Strong competition from smart-home ecosystems, hardware-based monitors and utility apps.
- Many companies offer similar services (differentiation is limited and rapid technological change increases the pressure to innovate

treat of new entry

- Time and cost of entry: basic energy tracking is rapid and cheap. However real-time IoT integration, machine learning and analytics require device testing and hardware compatibility work.
- Specialist knowledge requirements.
- Struggle with pricing for new entrants.
- Cost advatages of existing firms
- Customer trust, data privacy, hardware certification

threat of substitution

- The existing energy companies already providing the energy data to customers
- In-app consumption analytics reduces the need for a third application.

buyer power

- B2C: Homeowners using Sense devices and applications - various alternatives and low swiching costs.
- B2B: Energy companies adapting the Sense application into their platform

supplier power

- Hardware manufacturers: Comonents/specialized components.
- Cloud service dependence: Apps rely on major cloud providers for real-time processing and storage.

Sense Value Chain

Support Activities

Firm Infrastructure

The company's infrastructure integrates research and development, data analytics, and strategic partnerships to facilitate efficient production, cloud services, and operations.

Human Resource Management

Sense's human resource management emphasizes hiring skilled engineers, data scientists, and support staff, while fostering innovation, and continuous learning within the organization.

Technology Development

Sense's technology focuses on enhancing machine learning, NILM algorithms, and hardware design to enhance energy detection accuracy and overall system performance.

Procurement

Sense's buying process is about getting top-quality electronic parts, sensors, and cloud services. This helps make sure the hardware works well and data flows smoothly.

Primary Activities

Inbound Logistics

The inbound logistics of Sense is responsible for obtaining the necessary materials for device manufacturing. This includes finding, moving, and keeping them stored safely.

Operations

Sense's operations focus on assembling energy monitors, integrating software systems, and ensuring product quality and performance.

Outbound Logistics

Sense's outbound logistics handles the distribution of finished energy monitors to consumers, retailers, and utility partners.

Marketing and Sales

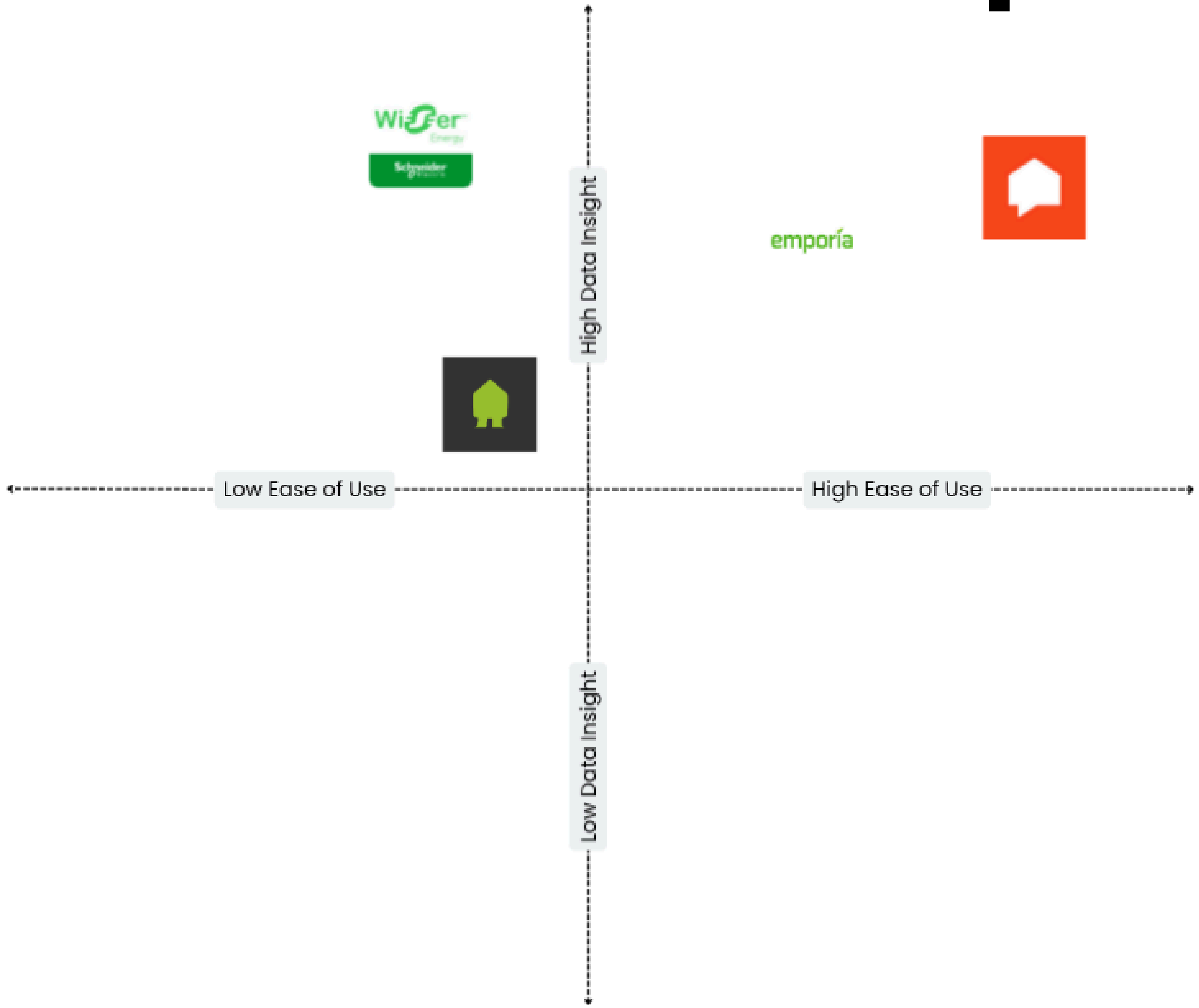
Sense's marketing and sales focus on promoting energy efficiency through digital campaigns, partnerships with utilities, and direct consumer sales.

Service

Sense's service provides customer support, app updates, and ongoing maintenance to ensure accurate energy monitoring and user satisfaction.

MARGIN

Position Map



VRIO Analysis for Sense

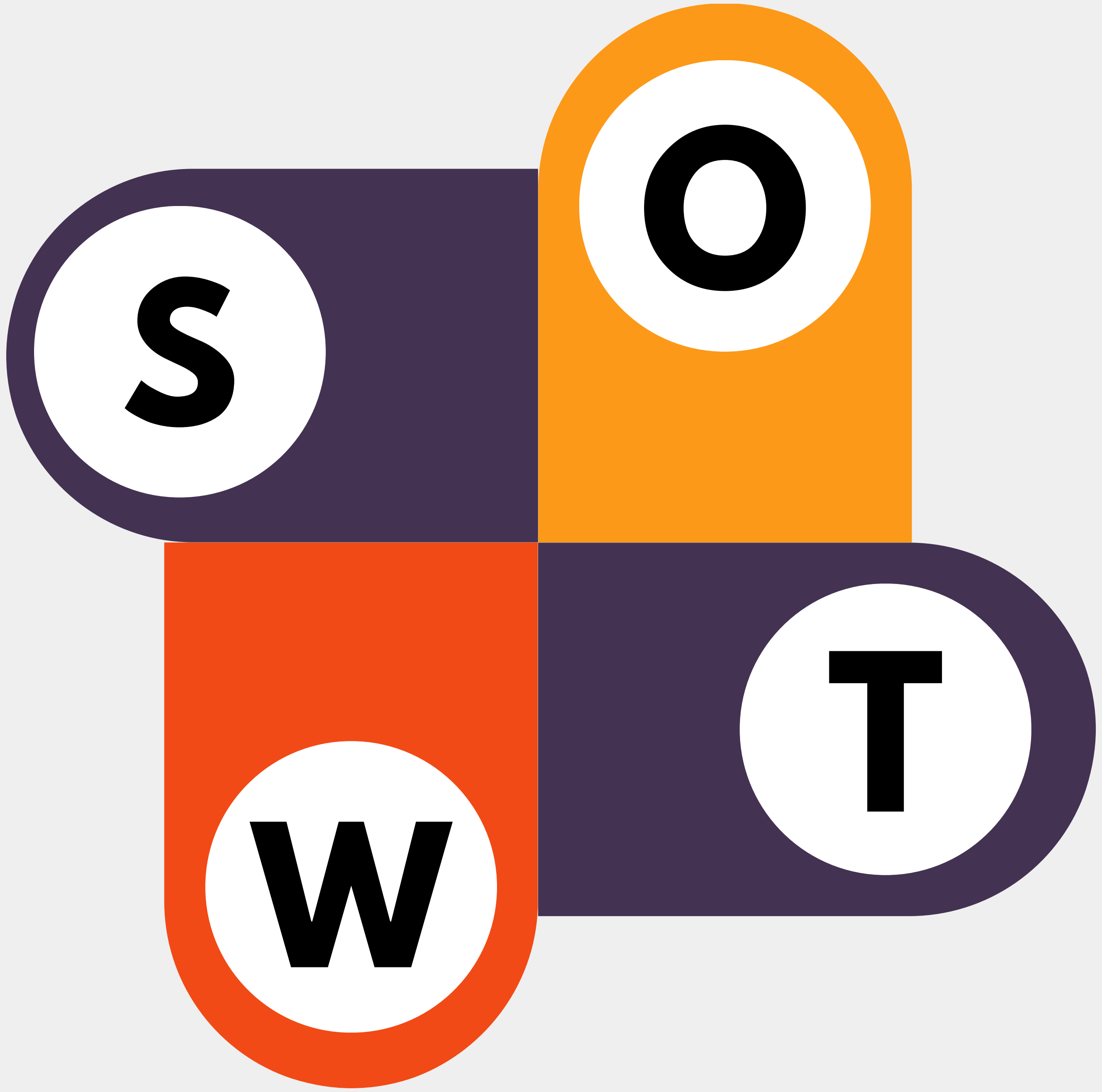
	Value	Rare	Inimitate	Organise
Embed with AI	✓	✓	✓	✓
Smart control	✗	✗		

Strengths

- AI-embedded
- Grid and home efficiency
- Empowering and engaging users

Weaknesses

- High Price
- Complex installation
- Device misidentification



Opportunities

- Rapid growth in electricity demand
- Infrastructure is not everything
- The need for the data-driven grid

Threats

- Intense Competition
- Data Privacy and Security Concerns
- Evolving Technology and Standards

Tows Analysis for Sense

TOWS	S	W
O	• use ML tech and partnership to embed in smart meters • bundle with EV charges and solar for a smart energy ecosysytem • promote “sense-enabled homes” for eco-conscious users	• use government and utility programs to susidize installation cost • develop localized models • improve ML accuracy by community data sharing
T	• with strong partnerships and funding to stay ahead technologically • highlight data security and transparency to ease concerns	• reducing installation camplexity through OEM partnerships • maintain "front office" to protect brand identity • strengthen trust through transparent privacy

Weak Point 1

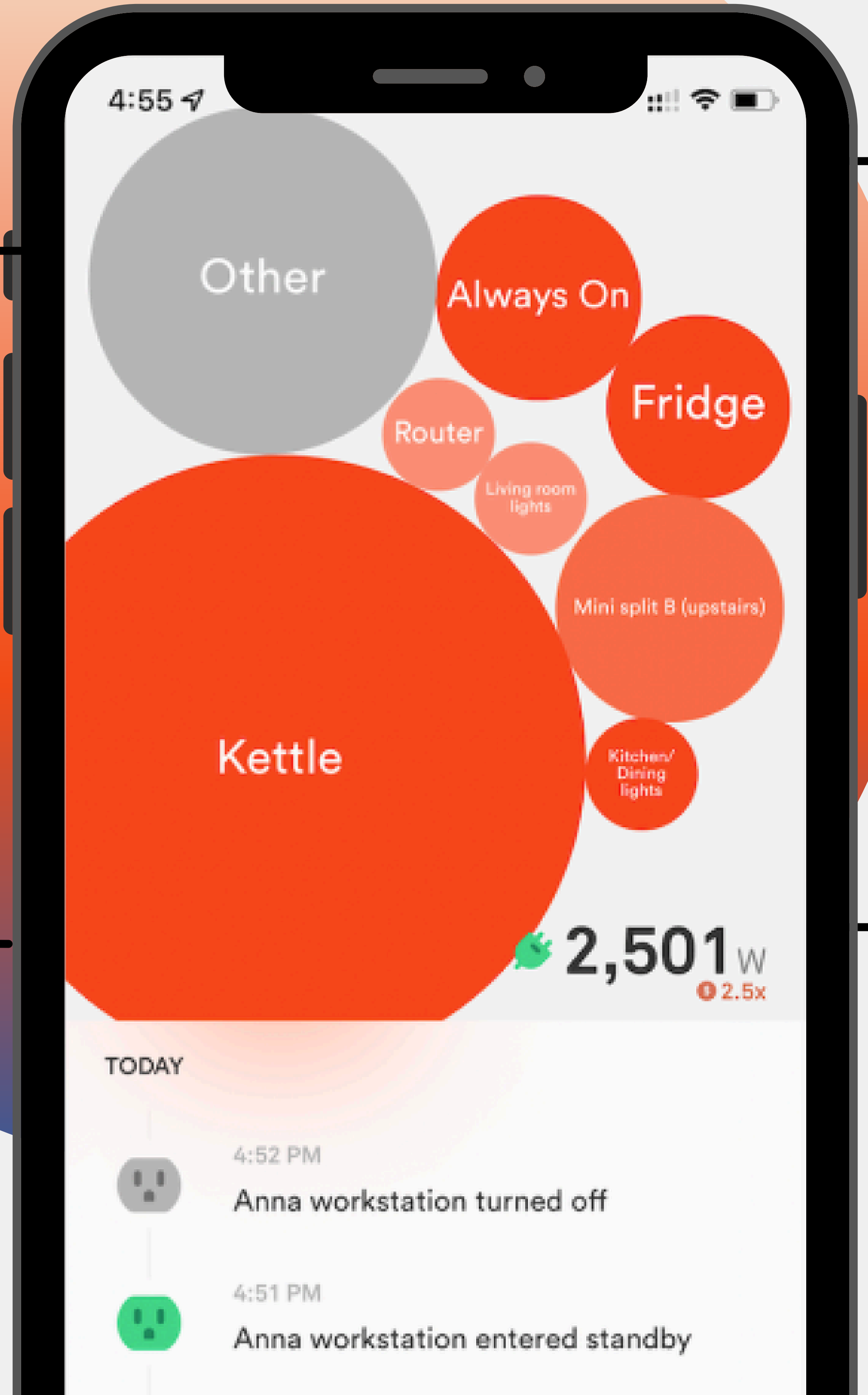
Limited device recognition accuracy

Problem: The AI often fails to detect all appliances, or takes weeks/months to learn them.
Root cause: Not enough labeled data diversity and signal overlap between similar devices.

Weak Point 2

Installation complexity & cost

Problem: Requires a licensed electrician to install; deters mainstream adoption.



Weak Point 3

User engagement & perceived value plateau

Problem: Users often lose interest after initial curiosity if insights don't change behavior.

Weak Point 4

Limited European adoption / compatibility

Problem: The electrical system and grid architecture differ (e.g., 230 V single phase vs. U.S. split phase).

Solutions

Limited device recognition accuracy

- Hybrid detection approach: Combine signature-based Machine Learning with contextual data inputs (smart plugs, Wi-Fi IoT devices).
- User-driven labeling: Allow users to manually label energy events.
- Community signature sharing: Build an anonymized global "device fingerprint" database that improves identification across homes.

User engagement & perceived value plateau

- Gamify energy savings: Points, challenges, "energy efficiency streaks" comparisons with similar homes.
- Personalized notifications: Detect unusual activity patterns.
- Seasonal insights: Push relevant tips.

Installation complexity & cost

- Partner with utility installers: Bundle the Sense monitor with smart meter upgrades, offering free installation or rebate-based installation credits.
- DIY-ready version: Develop a simplified "plug-in" version for homes/apartments using smart plugs or subpanels.
- Installation support app: Add AR-assisted step-by-step installation guidance through the app.

Limited European adoption/compatibility

- Localized hardware SKUs: Design models compatible with European distribution boards and DIN-rail standards.
- Multilingual support & regional partnerships: Distribute via major energy retailers in Europe (e.g., Iberdrola, EDF, E.ON).
- Localized analytics: Adapt algorithms for EU appliance types and usage patterns.

End

Thank you

Do you have any questions?